

# A Finite-Dimensional Obstruction Program for Schauder Reorderings of Quasi-Greedy Markushevich Bases

Automated mathematics run `fa_banach_001`, lane 11

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**Status.** Candidate partial result, likely valid, subject to human review. This packet proves a finite-dimensional reduction; it does not solve the open Banach-space problem.

## 1 Source question

Miguel Berasategui asks in the introduction of [1] (arXiv:2510.06398v1, PDF page 2):

“It is an open problem whether every quasi-greedy Markushevich basis has a Schauder reordering.”

The paper works throughout with Banach spaces. The following image is a crop rendered from the original arXiv PDF.

While quasi-greediness was originally defined for Schauder bases, the study of Markushevich bases with this property began shortly afterwards in [28], when Wojtaszczyk proved that quasi-greedy Markushevich bases are those for which the partial sums in (1.1) are uniformly bounded [28, Theorem 1] (see also [6, Theorem 4.1], where it is shown that even the Markushevich hypothesis is not necessary and follows from a pointwise boundedness hypothesis on the sums). This mirrors the classical equivalence for Schauder bases, though with the crucial difference that for quasi-greedy Markushevich bases, the order of the sums is not fixed, but depends on each  $f \in \mathbb{X}$  (the order may be undetermined by the TGA because multiple coefficients may have the same modulus, but the aforementioned equivalence holds for every order that meets the criteria [28, Theorem 1], [26, Theorem 2.1], [6, Theorem 4.1]).

It is an open problem whether every quasi-greedy Markushevich basis has a Schauder reordering (see [6, Problem 13.6], [7, Problem 14]), and much of the research on quasi-greedy bases focuses on Schauder bases, but the study of quasi-greedy bases without a Schauder hypothesis has continued since [28] (see for example [2, 6, 20, 16, 26], among others). In this context, it is a matter of interest to study which salient properties of Schauder bases are also present in quasi-greedy Markushevich bases.

The more recent quasi-Banach counterexample of Albiac–Ansorena–Berasategui [2] explicitly says that this question remains open under the Banach-space restriction.

## 2 Definitions

Let  $E$  be a  $d$ -dimensional Banach space with a normalized basis  $\mathcal{B} = (e_i)_{i=1}^d$  and biorthogonal functionals  $(e_i^*)_{i=1}^d$ . For  $A \subset \{1, \dots, d\}$  write

$$P_A x = \sum_{i \in A} e_i^*(x) e_i.$$

A finite set  $A$  is *greedy* for  $x$  if

$$|e_i^*(x)| \geq |e_k^*(x)| \quad (i \in A, k \notin A).$$

The basis is  $K$ -quasi-greedy if  $\|P_A x\| \leq K\|x\|$  for every  $x \in E$  and every greedy set  $A$ .

Define the *best ordering constant*

$$\beta(\mathcal{B}) := \min_{\sigma \in S_d} \max_{1 \leq m \leq d} \|P_{\{\sigma(1), \dots, \sigma(m)\}}\|.$$

Thus  $\beta(\mathcal{B})$  measures the smallest basis constant obtainable by reordering the finite basis.

### 3 Proof intuition

Greedy selection is local with respect to an  $\ell_2$ -sum: a global greedy set meets every finite block in a greedy set. Consequently uniform quasi-greediness of the blocks passes to their union without any loss in the constant. Reordering behaves in the opposite direction. Every ordering of the union induces an ordering on each block, and a global prefix realizes each induced block prefix after restriction to that block. Therefore a block whose best ordering constant is large forces a large global partial-sum projection in every ordering. A sequence of such blocks gives the desired infinite obstruction.

### 4 Finite-dimensional block reduction

**Theorem 1** (Block obstruction). *Fix  $K, M \geq 1$ . For each  $j \in \mathbb{N}$ , let  $E_j$  be a finite-dimensional Banach space with a normalized basis  $\mathcal{B}_j = (e_{j,r})_{r=1}^{d_j}$  and biorthogonal functionals  $(e_{j,r}^*)_{r=1}^{d_j}$ . Assume*

1.  $\sup_{j,r} \|e_{j,r}^*\| \leq M$ ;
2. every  $\mathcal{B}_j$  is  $K$ -quasi-greedy; and
3.  $\beta(\mathcal{B}_j) \rightarrow \infty$ .

Then, in the Banach space

$$X = \left( \bigoplus_{j=1}^{\infty} E_j \right)_{\ell_2},$$

the union  $\mathcal{B} = \bigcup_j \mathcal{B}_j$ , in any enumeration, is a normalized,  $M$ -bounded,  $K$ -quasi-greedy Markushevich basis. No permutation of  $\mathcal{B}$  is a Schauder basis of  $X$ .

*Proof.* Embed each  $e_{j,r}$  in the  $j$ th coordinate of  $X$ , and similarly embed  $e_{j,r}^*$  in  $X^* = (\bigoplus_j E_j^*)_{\ell_2}$ . The resulting systems are biorthogonal. Finite block-supported vectors are dense in  $X$ , and if all embedded coordinate functionals vanish at  $x = (x_j)_j$ , then every  $x_j = 0$ . Thus the union is a Markushevich basis. It is normalized and  $M$ -bounded by hypothesis.

Let  $x = (x_j)_j \in X$ , and let  $A$  be a finite greedy set for  $x$  relative to the union. Put  $A_j = A \cap \{(j, r) : 1 \leq r \leq d_j\}$ . If  $A_j$  is nonempty, then for every selected  $r$  and every unselected  $s$  in the same block, the global greedy inequality gives

$$|e_{j,r}^*(x_j)| \geq |e_{j,s}^*(x_j)|.$$

Hence  $A_j$  is a greedy set for  $x_j$  in  $\mathcal{B}_j$ ; the empty case contributes zero. Therefore

$$\|P_A x\|_X^2 = \sum_j \|P_{A_j} x_j\|_{E_j}^2 \leq K^2 \sum_j \|x_j\|_{E_j}^2 = K^2 \|x\|_X^2.$$

This proves that the union is  $K$ -quasi-greedy.

Now fix an arbitrary permutation  $\pi$  of the union. Its restriction to the  $j$ th block induces a permutation  $\sigma_j \in S_{d_j}$ . By the definition of  $\beta(\mathcal{B}_j)$ , there exists  $m_j \in \{1, \dots, d_j\}$  such that

$$\left\| P_{\{\sigma_j(1), \dots, \sigma_j(m_j)\}} \right\| \geq \beta(\mathcal{B}_j).$$

Let  $N_j$  be the position in the global order  $\pi$  of the coordinate  $e_{j, \sigma_j(m_j)}$ . At time  $N_j$ , precisely the first  $m_j$  coordinates of the induced order  $\sigma_j$  have appeared from block  $j$ . Consequently, the global initial projection  $S_{N_j}^\pi$ , restricted to the isometric copy of  $E_j$ , is exactly the displayed block projection. Hence

$$\|S_{N_j}^\pi\| \geq \beta(\mathcal{B}_j).$$

Because  $\beta(\mathcal{B}_j) \rightarrow \infty$ , the initial projections associated with  $\pi$  are not uniformly bounded. Thus the union is not a Schauder basis in the order  $\pi$ . Since  $\pi$  was arbitrary, no Schauder reordering exists.  $\square$

**Corollary 2** (Necessary finite-dimensional consequence of a positive answer). *If every quasi-greedy Markushevich basis in a Banach space admits a Schauder reordering, then for every  $K, M \geq 1$  there is  $C(K, M) < \infty$  such that*

$$\beta(\mathcal{B}) \leq C(K, M)$$

*for every normalized finite-dimensional  $K$ -quasi-greedy basis whose coordinate functionals have norm at most  $M$ .*

*Proof.* If no such  $C(K, M)$  existed, choose one finite basis for each integer  $j$  with  $\beta(\mathcal{B}_j) \geq j$ . The theorem would then give a quasi-greedy Markushevich basis in a Banach space with no Schauder reordering.  $\square$

## 5 Verification notes

The proof has two load-bearing points. First, a global greedy set really does intersect each block in a block-greedy set; no comparison with coefficients in other blocks is needed after this observation. Second, at the global time when the  $m_j$ th coordinate of a block appears, restriction of the global prefix projection to  $E_j$  is exactly the first- $m_j$  projection in the induced block order. Both are exact identities, not asymptotic estimates.

The uniform bound on the coordinate functionals prevents coefficient pathologies across the direct sum and places the resulting system in the usual semi-normalized,  $M$ -bounded framework. No computational verification is needed. The separate file `verification.md` records an adversarial check of the argument.

## 6 Limitations and novelty status

This is a reduction, not a construction of the finite-dimensional blocks. The existence of a family satisfying the three hypotheses is open here. No converse is claimed: a dimension-free best-ordering bound for all finite bases does not immediately produce a coherent order of an arbitrary infinite basis.

For the bounded novelty check on 9 August 2026, the run registry and arXiv were searched using combinations of the phrases *finite-dimensional quasi-greedy basis*, *permutation*, *best ordering constant*, *Schauder reordering*, and *direct sum*, together with arXiv ids 2510.06398 and 2510.13693. No exact occurrence of the theorem above was found. The later paper [2] proves a quasi-Banach counterexample but explicitly states that the Banach restriction remains open. Novelty of this elementary reduction is plausible, not certified; it may be folklore.

## 7 Human-review recommendation

Review as a likely valid partial result. The mathematical proof is short; specialist effort is best spent on (i) checking whether the reduction is already standard, and (ii) testing known finite-dimensional conditional quasi-greedy constructions for divergence of  $\beta$  under uniform  $K$  and  $M$ .

## References

- [1] M. Berasategui, *Quasi-greedy Markushevich bases, duality and norming subspaces*, arXiv:2510.06398v1 (2025).
- [2] F. Albiac, J. L. Ansorena, and M. Berasategui, *When Greedy Approximation Breaks: Counterexamples in Quasi-Banach Spaces*, arXiv:2510.13693 (2025).